

“THE **LIGHT** WHICH WE RECEIVE FROM THE **CLEAR SKY** IS DUE... TO **SMALL SUSPENDED PARTICLES** WHICH **DIVERT** THE LIGHT FROM ITS **REGULAR COURSE**. ”

Lord Rayleigh, English scientist, from *On the Light from the Sky, its Polarization and Color*, 1871

The sky appears blue because of Rayleigh scattering—the scattering of sunlight by molecules in the air.

AROUND 1870, ITALIAN PHYSICIAN CAMILLO GOLGI

(1843–1926) developed a technique to stain brain and other tissue to be able to observe them under the microscope. He used this method to identify **neurons** (nerve cells) which process and transmit information between the brain and other

parts of the body. For his work in this field, Golgi was awarded the Nobel Prize in Physiology or Medicine in 1906.

In a related discovery, German scientists **Gustav Theodor Fritsch** (1838–1927) and **Eduard Hitzig** (1839–1907) demonstrated the link between **electricity** and **brain function**. They showed

that an electric stimulus applied to different parts of a dog's brain would cause distinct muscular contractions.

The same year, English mathematician **William Kingdon Clifford** (1845–79) suggested that energy and matter are caused by the **curvature of space**. Clifford died young, and was unable to

develop the theory further; his ideas resurfaced in German-born physicist Albert Einstein's general theory of relativity (see 1914–15).

British astronomer **Joseph Norman Lockyer** (1836–1920) and his French colleague **Pierre Jules César Janssen** (1824–1907) independently suggested that certain lines in the Sun's spectrum were produced by a previously unknown element (see 1868–69). In 1870, Lockyer named this element **helium**, after the Greek Sun god—Helios.

Also in 1870, French microbiologist **Louis Pasteur** (1822–95) published a book documenting the mysterious disease killing silkworms, tracing it back to microbes. This, together with the discovery of the anthrax bacterium (see 1876–77) led to the development of **germ theory**.

In 1871, English scientist **Lord Rayleigh** (1842–1919) discovered that when light bounces off small particles, it scatters—now called **Rayleigh scattering**. He stated that for visible light to scatter, the particles must be smaller than 400 to 700 nanometers, that is, smaller than the wavelength of the light being scattered.

Ape and man

This illustration from an 1863 book by English biologist Thomas Henry Huxley—an advocate of Charles Darwin's theories—shows similarities between humans and living apes.



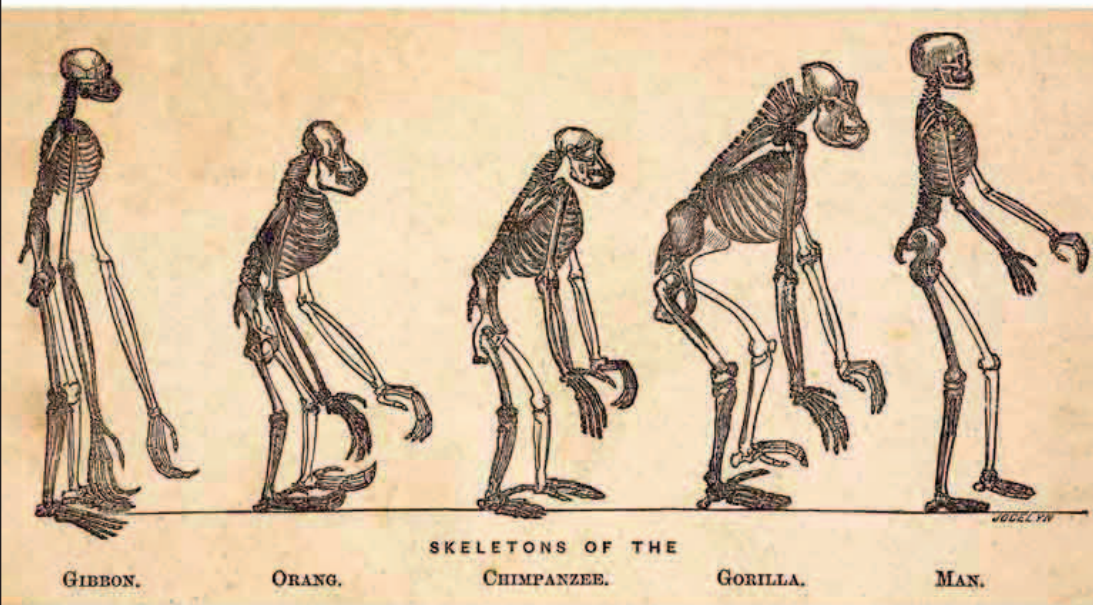
LOUIS PASTEUR
(1822–95)

While working at Lille University, France, Louis Pasteur investigated the problem of beer and wine (and later milk) going sour. He found that this was caused by bacteria, which could be killed by boiling (pasteurization). These studies helped Pasteur develop the germ theory of disease, and the prevention of disease by vaccination.

This year, English naturalist **Charles Darwin** published a book on **human evolution**—*The Descent of Man, and Selection in Relation to Sex*. He had been wary of publishing on the topic before, expecting the sort of furor that had accompanied the publication of *On the Origin of Species* in 1859.

“WITH **SAVAGES**, THE **WEAK** IN BODY OR MIND ARE **SOON ELIMINATED**... WE **CIVILIZED MEN**, ON THE OTHER HAND, DO OUR **UTMOST** TO **CHECK THE PROCESS OF ELIMINATION**. ”

Charles Darwin, British naturalist, from *The Descent of Man*, 1871



February 21, 1870
William Kingdon Clifford
presents his paper on the
Space Theory of Matter

c.1870 Camillo
Golgi identifies
neurons in the brain

c.1870 British scientist
William Crookes uses
improved vacuum pump
technology to develop precursors
of Crookes tubes

March 1871 Charles
Darwin publishes
*The Descent of Man, and
Selection in Relation to Sex*

1870 Gustav Theodor
Fritsch and Eduard
Hitzig demonstrate
the link between
electricity and
brain function

1870 Norman Lockyer
and Jules Janssen
identify **helium** in the
solar spectrum

1870 Louis Pasteur
publishes *Studies on
Silk Worm Disease*

1871 Lord Rayleigh explains
why the sky is blue—
Rayleigh scattering

1871 Swiss biochemist
Johann Friedrich Miescher
publishes a paper describing
his discovery of **nuclein**
from white blood cells